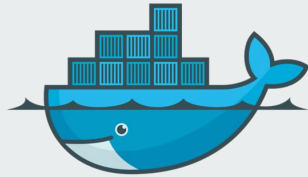




Docker & Kubernetes



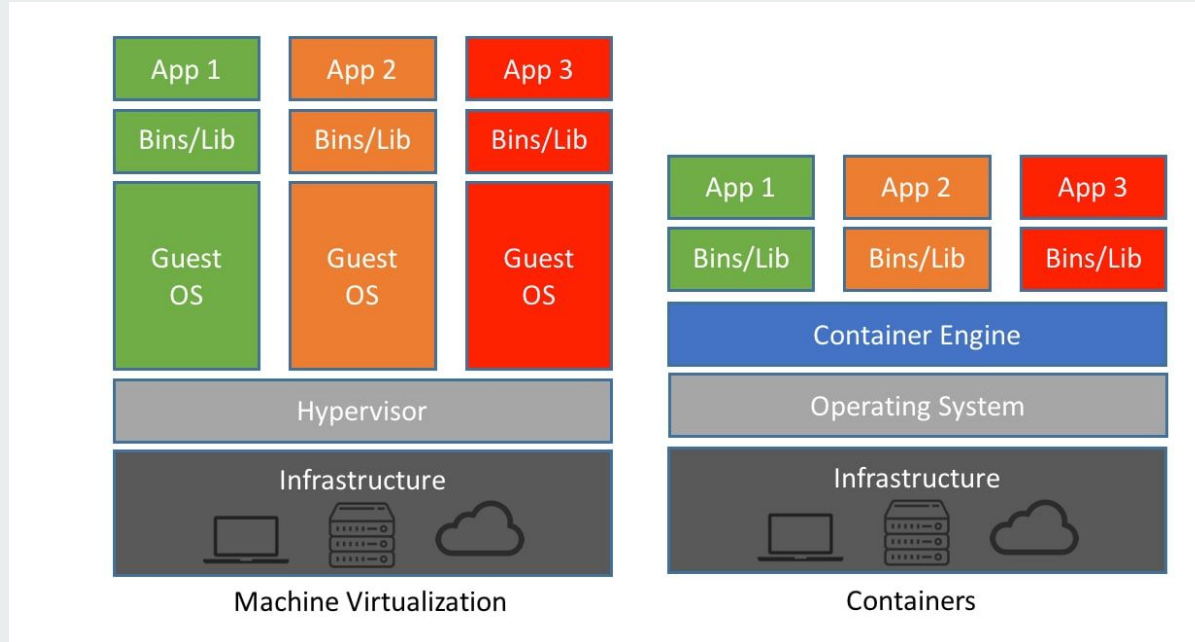
docker



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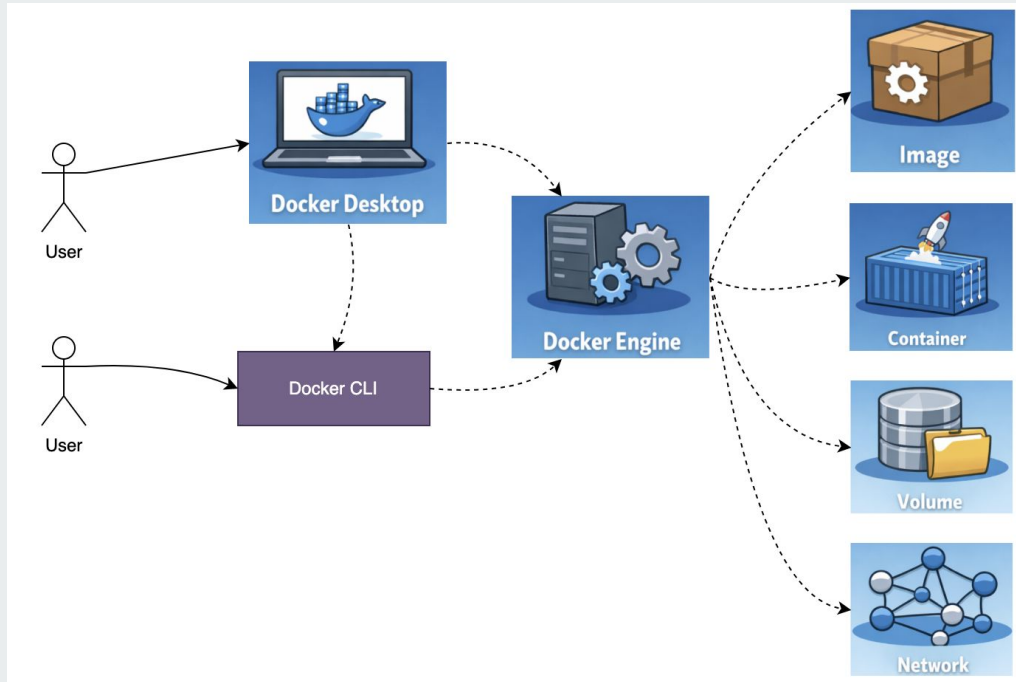
Based on course materials and lectures by Professor Mohammad & SDL

Why Docker (When Virtualisation Is Not Enough?)



Source: <https://www.netapp.com/blog/containers-vs-vms/>

Docker Platform and Core Objects



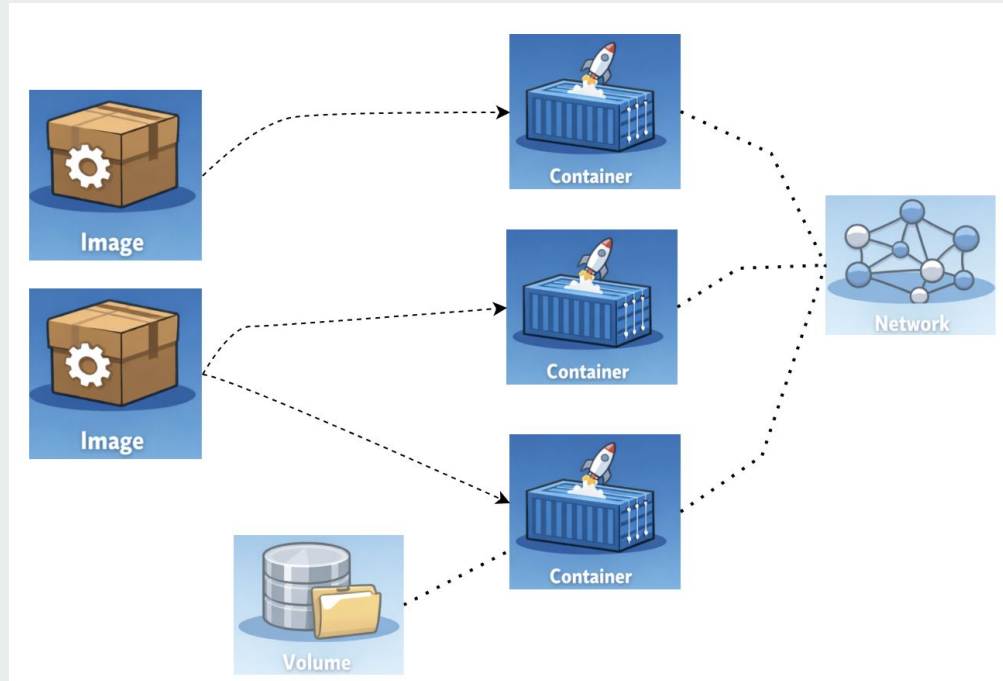
`docker image ls`

`docker container ls`

`docker volume ls`

`docker network ls`

Docker Core Objects



Image

A template or blueprint

Container

A temporary running instance of an image

Volume

Persistent storage mounted to a container

Network

Connects containers

Docker cheat sheet



Three main components of Docker

- Docker Client
- Docker Daemon
- Docker Registry

Usually most containers are Linux-based

- Windows: WSL
- Mac: Lightweight Linux VM of Docker

```
docker run -e KEY=value <image>
docker run -it <image> bash
docker run -d -p 8001:8001 --name svr_nginx
```

```
docker restart <container>
docker pause <container>
docker unpause <container>
```

```
docker ps
docker top <container>
docker inspect <image>
docker history image1
```

```
docker rm -f $(docker ps -aq)
docker system prune -a
```

https://docs.docker.com/get-started/docker_cheatsheet.pdf

Dockerfile & docker-compose.yml

Dockerfile

Dockerfile

```
docker build -t cars-db ./db
```

```
docker build -t cars-cli ./cli
```

```
docker network create
```

```
docker volume create
```

```
docker run --rm -it \ \ --network ... -v $(pwd)/db:/db \ cars-cli
```

docker-compose.yml

```
docker compose up
```

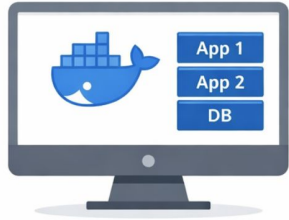
```
docker compose run --rm cli
```

vs

Kubernetes (High-level Overview)

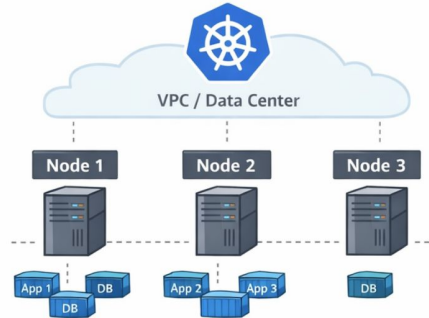
Docker

Single Machine



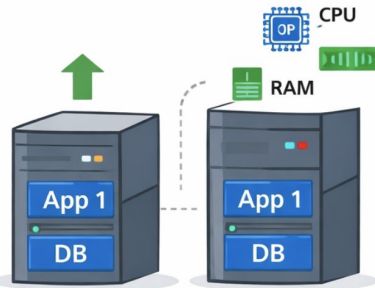
Kubernetes

Cluster of Machines



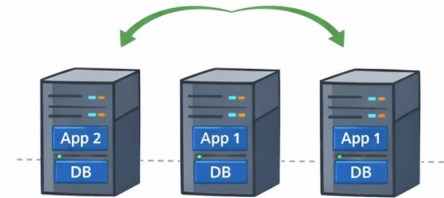
Scale Up

Vertical Scaling

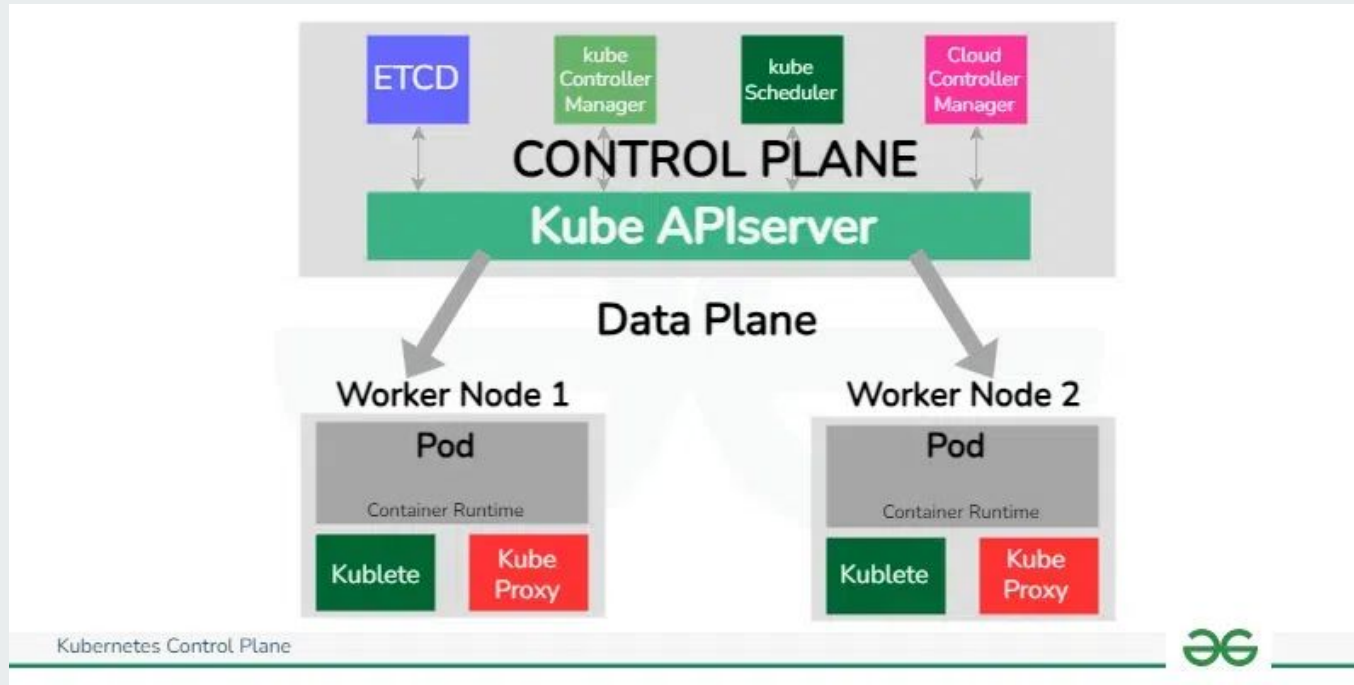


Scale Out

Horizontal Scaling

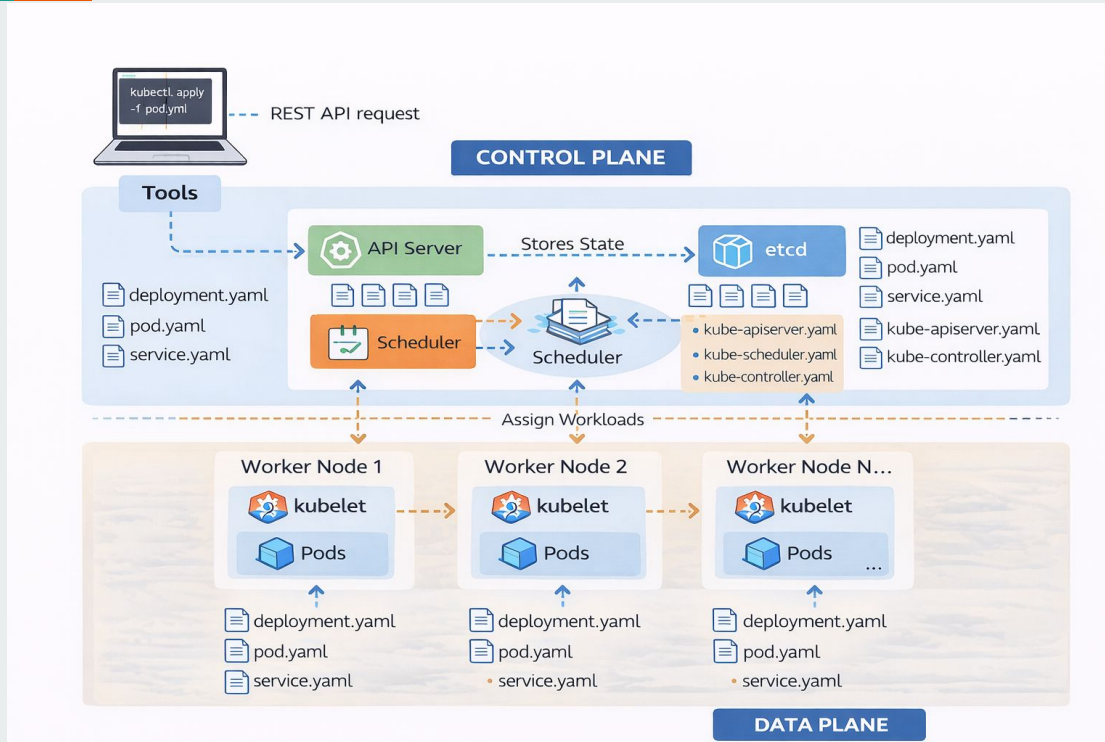


Kubernetes (Components and Architecture)



Source: <https://www.geeksforgeeks.org/devops/what-is-kubernetes-control-plane/>

Kubernetes (Components and Architecture)





Q & A



Thank you !!!

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